

Ground Control Station Interface Control Document (ICD)

Project: Tactical Overwatch Module for Tracking, Observation, and Mapping (TOMTOM)

Document Number: GCS-ICD-001

Prepared By: Serena Green

Version: 001

1. Purpose

The purpose of this Interface Control Document (ICD) is to define, document, and control the

interfaces associated with the Ground Control Station (GCS) subsystem of the Tactical Overwatch Module for Tracking, Observation, and Mapping (TOMTOM). The document establishes the physical, electrical, data, software, and human interfaces required to support successful integration, operation, maintenance, and future expansion of the Ground Control Station throughout the project lifecycle.

The Ground Control Station serves as the primary command-and-control subsystem for TOMTOM, providing the operator with the capability to plan missions, configure vehicle parameters, monitor aircraft health and telemetry, observe payload data, issue flight commands, and record mission information. Successful operation of these functions depends upon reliable communication and interaction between multiple hardware and software components. This document defines those interfaces to ensure compatibility between interconnected subsystems and to minimize integration risk during assembly, verification, and operational testing.

The ICD provides a controlled engineering baseline that identifies how information, electrical power, and operator interactions are exchanged within the Ground Control Station and between the Ground Control Station and external systems. By documenting interface characteristics, responsibilities, and verification methods, this document supports configuration management while providing a common technical reference for design implementation, integration activities, verification planning, and future system modifications.

This document also establishes traceability between the Ground Control Station Concept of Operations, Ground Control Station Requirements Specification, engineering trade studies, verification activities, and system architecture models. As the Ground Control Station evolves through prototype development and subsystem testing, this ICD will be updated under configuration control to accurately reflect the implemented interface architecture and maintain consistency across all project documentation.

2. Scope

This Interface Control Document (ICD) applies exclusively to the Ground Control Station (GCS) subsystem of the Tactical Overwatch Module for Tracking, Observation, and Mapping (TOMTOM). The document defines the interfaces required to support communication, power distribution, data exchange, software interaction, and operator control between Ground Control Station components and external systems. These interface definitions establish the technical baseline necessary to support subsystem integration, verification, maintenance, and future system expansion.

The scope of this document includes all physical, electrical, data, software, and human interfaces that cross subsystem boundaries within the Ground Control Station architecture. Interface characteristics such as connectors, communication protocols, signal direction, interface purpose, operational responsibilities, and verification methods are documented to ensure compatibility

between interconnected hardware and software components. Where applicable, this document also identifies the verification activities used to demonstrate proper interface implementation during subsystem testing and integration.

This document does not define the internal design of individual hardware components, software implementation details, aircraft flight control algorithms, payload processing, or system-level operational behavior. Those subjects are addressed within the Ground Control Station Concept of Operations, Ground Control Station Requirements Specification, Ground Control Station Trade Studies, and the TOMTOM system documentation.

The interface definitions contained within this document represent the current Ground Control Station prototype configuration and provide the controlled engineering baseline from which future design modifications, interface updates, and verification activities will be managed under project configuration control.

3. System Overview

The Ground Control Station (GCS) is a portable command-and-control subsystem designed to support planning, execution, monitoring, and post-mission activities for the Tactical Overwatch Module for Tracking, Observation, and Mapping (TOMTOM). The system provides the primary interface between the operator and the unmanned aircraft by integrating mission software, communications hardware, power distribution, and operator controls into a transportable field workstation. The GCS enables the operator to configure missions, monitor aircraft and payload status, issue flight commands, receive telemetry, and record mission information throughout all phases of operation.

The current Ground Control Station prototype utilizes commercial off-the-shelf (COTS) hardware integrated within a ruggedized equipment case to provide a modular and maintainable architecture. The system was developed with an emphasis on portability, subsystem integration, and future expandability while minimizing unnecessary system complexity. The modular design allows individual hardware components to be replaced or upgraded with minimal impact to the remainder of the Ground Control Station architecture.

The Ground Control Station consists of several major subsystems that collectively support mission execution. The embedded computing platform serves as the primary processing element and hosts the Linux operating system, QGroundControl, and supporting mission software. The power subsystem provides portable electrical power through an internal rechargeable battery and distributes regulated power to connected hardware through a protected electrical distribution architecture. The communications subsystem enables bidirectional data exchange between the Ground Control Station and the aircraft using a telemetry radio and the MAVLink

communication protocol. The Human-Machine Interface (HMI) provides the operator with visual displays and input devices required for mission planning, command and control, and system monitoring. The mechanical enclosure protects internal hardware while providing a transportable platform suitable for laboratory and field operations.

The interfaces between these subsystems form the foundation of Ground Control Station operation. Physical interfaces provide mechanical and electrical connectivity between hardware components, electrical interfaces distribute power throughout the system, data interfaces exchange commands and telemetry between subsystems, and human interfaces enable effective interaction between the operator and mission software. These interfaces are defined in the following sections of this document to establish a controlled engineering baseline supporting subsystem integration, verification, and future development activities.

4. Interface Overview

The Ground Control Station (GCS) relies on a series of well-defined interfaces that enable the exchange of electrical power, digital information, operator commands, and mission data between internal subsystems and external systems. These interfaces provide the foundation for subsystem integration by establishing standardized methods of communication and interaction throughout the Ground Control Station architecture. Clearly defining each interface reduces integration risk, simplifies maintenance activities, and supports future subsystem expansion while maintaining compatibility with the overall TOMTOM system architecture.

For the purposes of this document, Ground Control Station interfaces are organized into four primary categories: physical interfaces, electrical interfaces, data interfaces, and human interfaces. Each category represents a unique method by which hardware, software, or the operator interacts with the Ground Control Station. Although these interfaces perform different functions, they operate together to provide a cohesive command-and-control capability supporting all phases of mission execution.

Physical interfaces define the mechanical connections between hardware components and establish the physical pathways used to connect displays, peripheral devices, communication equipment, and power components. These interfaces include the connectors, cables, and attachment methods required to assemble and operate the Ground Control Station while supporting maintenance and future hardware upgrades.

Electrical interfaces describe the distribution of electrical power throughout the Ground Control Station. These interfaces identify the flow of power from the internal battery system through protective devices and voltage regulation hardware to the individual subsystems requiring electrical power. Proper definition of electrical interfaces supports safe operation, simplifies troubleshooting, and ensures compatibility between interconnected hardware components.

Data interfaces define the communication pathways used to exchange information between the Ground Control Station and external systems. These interfaces support the transfer of aircraft telemetry, operator commands, mission information, payload imagery, and system status data. Reliable implementation of these interfaces is essential to maintaining situational awareness and effective command and control during mission operations.

Human interfaces define the interaction between the operator and the Ground Control Station. These interfaces include the display, keyboard, pointing device, and controller used to monitor system status, configure missions, issue commands, and observe payload information. Together, these interfaces provide the operator with an intuitive and efficient method of controlling the Ground Control Station throughout all operational phases.

5. Physical Interfaces

Physical interfaces establish the mechanical and electrical connections required for operation of the Ground Control Station. These interfaces define the connectors, ports, and physical attachment methods used to interconnect hardware components while providing a reliable foundation for power distribution, communications, and operator interaction. Standardized physical interfaces simplify assembly, maintenance, subsystem replacement, and future hardware expansion by reducing the need for custom interface solutions.

The Ground Control Station utilizes commercially available interface standards whenever practical to improve compatibility, reduce integration complexity, and support long-term maintainability. Major physical interfaces include connections between the embedded computer and external display, USB peripheral devices, telemetry communications hardware, and the internal power subsystem. Selection of these interfaces was guided by the results of the Ground Control Station trade studies and the overall system architecture developed for the TOMTOM project.

Each physical interface documented within this section identifies the connected subsystems, interface purpose, connector type, signal direction, and verification method used to demonstrate successful integration during subsystem testing. These interface definitions establish the baseline configuration for the current Ground Control Station prototype while supporting future modifications under project configuration control.

GCS-INT-001

Embedded Computer ↔ Portable Display

Purpose

Provides visual presentation of mission software, telemetry, payload imagery, and operator interface.

Attribute	Description
Interface Type	Physical/Data
Connector	HDMI
Direction	Output
Connected Equipment	Portable Monitor
Verification	Demonstration

GCS-INT-002

Embedded Computer ↔ USB Hub

Purpose

Provides expansion capability for operator peripherals and future hardware.

Attribute	Description
Connector	USB 3.0
Direction	Bidirectional
Verification	Test

GCS-INT-003

USB Hub ↔ Operator Peripherals

Supports:

- Keyboard
- Mouse
- Controller

6. Electrical Interfaces

Electrical interfaces define the distribution of electrical power throughout the Ground Control Station and establish the connections required to safely operate all internal subsystems. These interfaces include the rechargeable battery, master power switch, fuse protection, voltage regulation hardware, and internal power distribution network. Together, these components provide regulated electrical power to the embedded computer, display, communications equipment, peripheral devices, and future subsystem expansions.

The Ground Control Station has been designed around a portable battery-powered architecture to support field operation independent of external electrical infrastructure. Electrical interfaces were selected to maximize reliability, simplify maintenance, and improve operator safety while

maintaining compatibility with commercially available components. Protective devices such as fuse protection and the master power switch reduce the risk of equipment damage while allowing rapid isolation of electrical power during maintenance or emergency situations.

Electrical interface definitions identify power sources, operating voltages, connector types, direction of power flow, and subsystem responsibilities. Verification of these interfaces will be accomplished through inspection, continuity testing, voltage measurements, and functional subsystem operation during Ground Control Station integration and verification testing.

GCS-INT-004

Battery Interface

Attribute	Description
Battery	12.8 V LiFePO ₄
Connector	XT60
Direction	Source
Verification	Inspection/Test

GCS-INT-005

Master Power Switch

Purpose:

Provides complete isolation of electrical power during transportation, maintenance, and emergency shutdown.

GCS-INT-006

Power Distribution

Purpose:

Distributes regulated electrical power to:

- Embedded computer
- Display
- USB hub
- Cooling fans
- Future payloads

Voltage regulation is provided through the integrated DC-DC buck/boost converter.

7. Data Interfaces

Data interfaces define the exchange of digital information between the Ground Control Station, the unmanned aircraft, and supporting software applications. These interfaces enable transmission of operator commands, reception of aircraft telemetry, display of payload imagery, exchange of mission information, and communication between software applications operating within the Ground Control Station.

Reliable data communication is essential to successful mission execution. During operation, the Ground Control Station continuously exchanges information with the aircraft using standardized communication protocols while simultaneously processing operator inputs, displaying mission status, and recording operational data. These interfaces support command and control, situational awareness, and mission management throughout all phases of operation.

The primary communication interface between the Ground Control Station and the aircraft utilizes the MAVLink protocol transmitted through the telemetry communication subsystem. Additional software interfaces support communication between the Linux operating system, mission software applications, and supporting utilities required for Ground Control Station operation. Data interface definitions identify communication protocols, direction of information flow, interface purpose, and verification methods used to demonstrate successful information exchange during subsystem testing and integration.

GCS-INT-007

Flight Controller Interface

Attribute	Description
Protocol	MAVLink
Direction	Bidirectional
Verification	Test

Purpose:

Exchange flight commands and telemetry.

GCS-INT-008

Telemetry Interface

Purpose:

Transmit

- Aircraft status
- GPS
- Battery
- Commands
- Mission updates

GCS-INT-009

Payload Video Interface

Purpose:

Receive payload imagery.

Direction:

Aircraft → GCS

Verification:

Demonstration

GCS-INT-010

Mission Software Interface

Purpose:

Provides communication between:

- Linux
- QGroundControl
- MAVLink
- Operator

8. Human Interfaces

Human interfaces define the methods by which the operator interacts with the Ground Control Station during mission preparation, execution, recovery, and maintenance activities. These interfaces include the visual display, keyboard, pointing device, and optional game controller that collectively provide the operator with the ability to configure missions, monitor aircraft status, issue commands, and observe payload information.

The Human-Machine Interface (HMI) was developed to provide an intuitive and efficient operating environment while minimizing unnecessary operator workload during mission execution. The selected interface architecture supports simultaneous monitoring of aircraft telemetry, mission progress, payload imagery, and system health while allowing rapid access to mission planning and command functions. Interface design emphasizes usability, maintainability, and compatibility with the selected Ground Control Station hardware configuration.

Each human interface identified within this section defines the purpose of the interface, the operator functions it supports, and the verification activities required to demonstrate successful operation. Verification will include inspection of installed hardware together with functional demonstrations confirming that the operator can effectively interact with the Ground Control Station throughout all operational modes.

Operator Display

Displays

- Maps
- Telemetry
- Payload
- Vehicle status
- Alerts

Keyboard

Supports

- Mission planning
- Software configuration
- Data entry

Mouse / Touchpad

Supports

- Menu navigation
- Mission editing
- Interface interaction

Game Controller

Supports manual vehicle control when required.

9. Interface Requirements

The interface requirements contained within this document establish the minimum performance and compatibility characteristics required to support successful integration of the Ground Control Station subsystem. These requirements supplement the Ground Control Station Requirements Specification by focusing specifically on subsystem interfaces rather than overall system functionality. Collectively, they define the expected behavior of the physical, electrical, data, and human interfaces identified throughout this document.

Each interface requirement is intended to be measurable and verifiable through inspection, demonstration, or functional testing. Compliance with these requirements provides confidence that subsystem interfaces have been correctly implemented and are capable of supporting reliable Ground Control Station operation. Verification activities associated with these requirements are identified within the Ground Control Station Verification Matrix and corresponding test procedures.

GCS-INT-REQ-001

The Ground Control Station shall provide bidirectional communication with the aircraft.

GCS-INT-REQ-002

The Ground Control Station shall distribute regulated electrical power to internal subsystems.

GCS-INT-REQ-003

The Ground Control Station shall support operator interaction through keyboard and pointing-device input.

GCS-INT-REQ-004

The Ground Control Station shall display telemetry and payload information simultaneously.

GCS-INT-REQ-005

The Ground Control Station shall support future USB peripheral expansion.

10. Interface Verification

The interfaces defined within this Interface Control Document shall be verified through inspection, demonstration, and functional testing performed during Ground Control Station integration and subsystem verification activities. Verification methods were selected based on the characteristics of each interface and the level of confidence required to demonstrate compliance with the design intent.

Physical interfaces, including mechanical connections, power connectors, and peripheral interfaces, will primarily be verified through inspection and functional demonstrations to confirm proper installation and operation. Electrical interfaces will be verified through inspection of wiring and electrical components in conjunction with functional testing to demonstrate correct voltage distribution, continuity, and subsystem operation.

Data interfaces, including communications between the Ground Control Station and the aircraft, will be verified during subsystem integration testing using established telemetry and command procedures. Successful verification requires demonstration that information is transmitted, received, processed, and displayed correctly while maintaining communication integrity throughout normal system operation.

Human interfaces will be verified through operator demonstrations to ensure mission software, displays, and operator controls function as intended while supporting efficient execution of mission tasks. Verification activities will confirm that the Ground Control Station provides the operator with the required situational awareness, command capability, and interface accessibility identified within the Ground Control Station Concept of Operations.

The verification activities identified within this document establish traceability between interface definitions, Ground Control Station requirements, and the corresponding test procedures. Successful completion of these verification activities demonstrates that interface implementation satisfies the operational requirements established for the Ground Control Station subsystem.

Table 10-1. Interface Verification Matrix

Interface ID	Interface Description	Verification Method	Verification Procedure	Verification Evidence
--------------	-----------------------	---------------------	------------------------	-----------------------

GCS-INT-001	Embedded Computer ↔ Portable Display	Demonstration	TP-011	Display operation, screenshots, operator verification
GCS-INT-002	Embedded Computer ↔ USB Hub	Test	TP-011	Peripheral recognition and operational testing
GCS-INT-003	USB Hub ↔ Operator Peripherals	Test	TP-011	Keyboard, mouse, and controller functionality
GCS-INT-004	Battery ↔ Power Distribution System	Inspection / Test	TP-002	Voltage measurements, continuity checks, battery operation
GCS-INT-005	Master Power Switch ↔ Electrical System	Inspection / Test	TP-002	Successful power isolation and startup/shutdown verification
GCS-INT-006	Power Distribution ↔ Internal Subsystems	Test	TP-011	Regulated power delivery to all connected subsystems
GCS-INT-007	Ground Control Station ↔ Flight Controller	Test	TP-003	Successful command transmission and telemetry reception
GCS-INT-008	Ground Control Station ↔ Telemetry System	Test	TP-004	Communication link establishment, telemetry monitoring, and recovery verification
GCS-INT-009	Ground Control Station ↔ Payload Video Stream	Demonstration	TP-011	Successful payload image display and operator observation

GCS-INT-010	Mission Software Interfaces	Test	TP-002	Successful operation of Linux, QGroundControl, and associated mission software
-------------	-----------------------------	------	--------	--

Verification Completion Criteria

Each interface shall be considered successfully verified when the following conditions have been satisfied:

- The physical interface has been inspected and confirmed to match the approved Ground Control Station configuration.
- Electrical interfaces provide stable power without abnormal operation or interruption.
- Data interfaces successfully exchange commands, telemetry, and mission information without communication failure during normal operating conditions.
- Human interfaces allow the operator to effectively monitor system status and perform required mission functions.
- Verification evidence has been recorded within the applicable Test Report and corresponding Verification Matrix.
- All discrepancies identified during verification have been resolved or documented through the project issue tracking process prior to subsystem acceptance.

References

Ground Control Station Concept of Operations

Ground Control Station Requirements Specification

Ground Control Station Trade Study TS-001

Ground Control Station Trade Study TS-002

Ground Control Station Trade Study TS-003

Ground Control Station Trade Study TS-004

Ground Control Station Trade Study TS-005

Ground Control Station Trade Study TS-006

Ground Control Station Verification Matrix

Ground Control Station Test Procedures

TOMTOM System Concept of Operations